

Numerical Analysis – Homework #1

Utah State University Fall 2026

1.Consider the function $f(x) = \sin x$ near $x = 0$.

- (a) Write the Taylor polynomials $P_1(x)$, $P_3(x)$, and $P_5(x)$.
- (b) Graph $P_1(x)$, $P_3(x)$, and $P_5(x)$ as well as $f(x)$ over $[0, 2\pi]$.
- (c) Compute $P_1(0.1)$, $P_3(0.1)$, and $P_5(0.1)$ and find the absolute and relative error.
- (d) Compute $P_1(0.5)$, $P_3(0.5)$, and $P_5(0.5)$ and find the absolute and relative error.
- (e) Compute $P_1(1)$, $P_3(1)$, and $P_5(1)$ and find the absolute and relative error.

2.Consider the function $f(x) = \ln(1 + x)$ near $x = 0$.

- (a) Write the Taylor polynomials $P_1(x)$, $P_2(x)$, and $P_3(x)$.
- (b) Graph $P_1(x)$, $P_2(x)$, and $P_3(x)$ as well as $f(x)$ over $[0, 2]$.
- (c) Compute $P_1(0.1)$, $P_2(0.1)$, and $P_3(0.1)$ and find the absolute and relative error.
- (d) Compute $P_1(0.5)$, $P_2(0.5)$, and $P_3(0.5)$ and find the absolute and relative error.
- (e) Compute $P_1(1)$, $P_2(1)$, and $P_3(1)$ and find the absolute and relative error.

3.Construct a Taylor polynomial approximation to $f(x) = e^x$ with $x_0 = 0$ that is accurate to within 10^{-3} on the interval $[0, 1]$. Find a value M such that

$$|f(x_1) - f(x_2)| \leq M|x_1 - x_2|$$

for all x_1, x_2 on the interval. (Use the Mean Value Theorem.)**4.**Find the Taylor polynomials of order 2 and order 3 around $x_0 = 1$ for

$$f(x) = x^3 + x.$$

5.

Find intervals containing solutions to the following equations:

- (a) $x - 3^{-x} = 0$
- (b) $4x^2 - e^x = 0$